PROJECT 1: BANK MARKETING

Predicting term-deposit subscription for campaign selection

1. Executive summary

This project develops a pre-call classification model to identify customers with a higher likelihood of subscribing to a term deposit. The supplied bank-additional-full dataset contains 41,188 records and 20 predictors plus the binary target. The outcome is imbalanced: 88.73% did not subscribe and 11.27% subscribed. HistGradientBoosting achieved the highest validation average precision (0.4928) among the three tested approaches and was selected for final evaluation. At the illustrative validation-selected threshold of 0.14, the held-out test recall was 64.22% and precision was 42.25%.

2. Data exploration and preparation

The data includes customer demographics (age, job, marital status, education), financial indicators (default, housing and personal loans), contact and campaign history, and quarterly/monthly economic indicators. There are 41,188 observations and 21 columns including y. The supplied CSV contains no blank/null values. Categorical values marked 'unknown' are retained as an explicit level because the data dictionary identifies them as a possible class of undisclosed information.

The duration field was excluded from operational features. The dataset documentation states that duration is measured during the last contact and is not known before a call; using it for campaign targeting would create target leakage. The pdays value 999 was retained because it means the customer was not previously contacted, not that the value is missing. Categorical variables were one-hot encoded; numeric variables were median-imputed and standardised inside a pipeline. A stratified 70%/15%/15% train/validation/test split was used.

3. Baseline and model comparison

The majority-class baseline predicts 'no' for every customer. Its accuracy is 88.73%, but its recall for subscribers is 0%, so accuracy alone is misleading. Models were ranked primarily by validation average precision because the positive class is uncommon and the business objective is to identify likely subscribers.

Validation rank/model	Average precision	ROC-AUC	Precision @ 0.5	Recall @ 0.5	F1 @ 0.5
1. HistGradientBoosting	0.4928	0.8127	0.7029	0.2414	0.3594
2. Random forest	0.4861	0.8000	0.4780	0.5618	0.5165
3. Logistic regression	0.4648	0.8021	0.3616	0.6379	0.4615

HistGradientBoosting ranked first on average precision and ROC-AUC. Random forest had a higher F1 at the default 0.5 cutoff, while logistic regression achieved the highest recall at 0.5. This illustrates that model ranking depends on the operational objective and threshold, not one metric alone. HistGradientBoosting was retained as the selected model because average precision was specified as the primary ranking metric.

4. Decision threshold and trade-off

The threshold was selected using validation data by minimising an illustrative cost function: 5 × false negatives + 1 × false positives. This assumes that missing a genuine subscriber is five times as costly as making an unnecessary call. The resulting threshold was 0.14. On validation data this selected 1,049 customers, with 444 true positives, 605 false positives

and 252 false negatives (precision 42.33%, recall 63.79%). This ratio is an assumption for demonstration; the bank should replace it with measured call costs, expected deposit value and campaign capacity.

5. Held-out test results

Measure	Test result
Selected model / threshold	HistGradientBoosting / 0.14
Accuracy	86.08%
Precision	42.25%
Recall	64.22%
F1	0.5097
ROC-AUC	0.8144
Average precision	0.4928
True negatives / false positives	4,872 / 611
False negatives / true positives	249 / 447
Illustrative cost (5×FN + FP)	1,856

The selected operating point detects 447 of the 696 subscribers in the test set (64.22% recall), while 611 non-subscribers are also selected for contact. Relative to a higher threshold, the lower threshold reduces missed subscribers but increases unnecessary calls. The threshold should therefore be adjusted to the bank's call capacity and actual economics.

6. Recommendation, limitations and conclusion

Use the model to rank customers by predicted probability and, where operationally necessary, apply a threshold chosen using real campaign costs. Monitor precision, recall, average precision and conversion rates after deployment. Validate on a later campaign period before operational use. The random split does not test temporal generalisation; customer behaviour and macroeconomic conditions may change. Historical targeting decisions may also create selection bias. Predictions describe patterns in the historical data and do not prove that a call causes a subscription.

Reproducibility: the accompanying Jupyter notebook includes data loading, exploration, preprocessing, baseline, model fitting, validation ranking, threshold sensitivity, test evaluation and visual diagnostics. Install the dependencies in requirements.txt and run all cells from the project folder.